Top-Level Module Boundary: `l2_bdi_cache_top.v`

CPU CORE

INSTRUCTION FETCH (I-PORT: 256b REFILL)

LOAD-STORE UNIT (D-PORT: 32b WISHBONE)

l1i_addr[31:0]

l1i_rdata[255:0]

cpu_d_bus[31:0]

L1 DATA CACHE

TAG DIRECTORY
64 Sets × 2 Ways

DATA STORAGE
64 Sets × 256-Bit Lines

l1d_addr[31:0]

l1d_wdata[255:0]

l1d_rdata[255:0]

UPSTREAM BUS ARBITER

PORT 0: INSTRUCTION
CPU 256b
Refill · Priority

PORT 1: DATA CACHE
L1 Line Refill & Eviction

TRANSACTION ROUTER
L2 Response Demultiplexing

l2_write_req

l2_wdata[255:0]

l2_req_addr[31:0]

l2_req_valid

l2_rdata[255:0]

l2_resp_valid

STLF HAZARD INTERLOCK

4-STAGE WRITE PIPELINE

BDI COMPRESSOR & DCB GATING

ADAPTIVE POWER GOVERNOR

fwd_valid

fwd_data[255:0]

refill_bus[255:0]

selection[1:0]

ADDRESS PARTITIONER

index[5:0]

16x SYNCHRONOUS TAG RAMs

read_tags[23:0]

PARALLEL TAG COMPARATORS

match_vector

DYNAMIC WAY-HALTING LOGIC

tag[31:11]

comp_payload[136:0]

7-BIT TREE-PLRU REPLACEMENT

plru_victim[2:0]

DUAL DIRTY & ATOMIC EVICTION

DUAL-SLOT ALLOCATOR

evict_bus[287:0]

index[5:0]

wren_A/B[7:0]

way_rden[7:0]

80x M9K CACHE DATA BRAM (160,768 BITS)

raw_sub_block[136:0]

DUAL-SLOT SUB-BLOCK SELECTOR MUX

sel_sub_block[136:0]

5-CYCLE VALID PIPELINE

valid

SINGLE-CYCLE BDI DECOMPRESSOR

decomp_line[255:0]

STLF BYPASS MULTIPLEXER

final_data[255:0]

DETERMINISTIC OUTPUT REGISTERS

DRAM CONTROLLER

Signal Nets: Address / Tag / Sub-Block · 256b Data / Evict / Refill · Clock / Write / PLRU · Valid / Hit / Ready · Selection / Gating

Pipeline Stages: Stage 1: Ingestion, STLF Interlock & BDI Compression (Cycles 1–4) · Stage 2: Storage Core & Tag Match ($> 55\%$ Area, Cycles 2–4) · Stage 3: 1-Cycle Decompression & Output (Cycle 5)